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1. Introduction

This report presents the final software design for WasteTrack SL, a Digital Public Good system developed to help address the serious waste management challenges facing Freetown, Sierra Leone.

WasteTrack SL is designed to support several United Nations Sustainable Development Goals, with primary focus on SDG 11: Sustainable Cities and Communities and SDG 12: Responsible Consumption and Production. The system specifically supports Target 11.6, which focuses on reducing the environmental impact of cities, especially through better municipal solid waste management. It also supports Target 12.4, which promotes the environmentally sound management of waste throughout its lifecycle, and Target 12.5, which aims to reduce waste generation through prevention, recycling, reuse, and responsible consumption.

In Sierra Leone, especially in urban areas such as Freetown, unmanaged solid waste remains one of the most visible and dangerous public health and environmental problems. Freetown generates an estimated 600 tonnes of solid waste per day, yet only about 40% is formally collected (World Bank, 2023). As a result, piles of uncollected waste often block drainage systems, leading to flooding during the rainy season. These conditions also contaminate water sources and create breeding grounds for disease vectors such as mosquitoes and cholera-causing bacteria.

Waste management in Freetown is often fragmented, underfunded, and poorly mapped. This means that city authorities do not always have access to real-time information needed to deploy trucks, monitor collection coverage, or identify illegal dumping hotspots. WasteTrack SL responds to this problem by providing a hybrid digital platform that combines a smartphone mobile application, a web-based administrative dashboard, SMS alerts, and USSD access through *111# for citizens without smartphones.

Through the platform, citizens can report illegal dumping, overflowing bins, and missed collections. They can also track waste collection trucks, receive SMS notifications about collection schedules, and access recycling education. At the same time, city authorities

and waste operators can use the system to monitor reports, assign trucks, manage collection routes, and generate useful data for SDG reporting.

1.1 Problem Definition

Freetown faces a solid waste management crisis caused by several structural gaps. WasteTrack SL is designed to address the following key problems:

Reporting Gap

Citizens currently have no simple and reliable way to report illegal dumping, overflowing bins, or missed collections to the Freetown City Council. Many complaints are made through phone calls or informal communication, but these reports are often not recorded, tracked, or resolved. As a result, waste hotspots can remain unattended for weeks or months, while the Council lacks verified data showing where the problems are most serious.

Operational Gap

Waste collection truck operators often lack digital route management tools. Routes are usually planned manually, drivers may receive verbal instructions, and supervisors may not have real-time visibility of truck locations, capacity levels, or progress. This leads to inefficient routing, fuel waste, repeated collection in some areas, and missed service in others, especially in informal settlements.

Data Gap

The Freetown City Council, the Ministry of Lands, Housing and the Environment, and development partners often lack real-time, verifiable data on waste collection coverage,

illegal dumping hotspots, recycling rates, and SDG progress. Current waste management data is usually based on field surveys, which are not frequent enough to support timely decisions. Without reliable data, it becomes difficult to plan interventions, allocate resources, and report progress toward SDG 11 and SDG 12.

1.2 Target Users

WasteTrack SL serves four main user groups:

Citizens

Citizens include residents of Freetown communities such as Kissy, Lumley, Congo Town, Wilberforce, and surrounding areas. Both smartphone users and basic phone users are included. Citizens can use the mobile app or USSD *111# to report waste problems, check collection schedules, track trucks, receive alerts, and learn about recycling.

Waste Collection Drivers

Waste collection drivers are operators of trucks used by the Freetown City Council and contracted waste management companies. Drivers use the mobile application to receive route instructions, mark completed collection points, update truck capacity, and report access or safety challenges during collection.

City Council Officers / Waste Supervisors

These are officers working with the Freetown City Council Waste Management Unit. They use the web dashboard to monitor waste reports, track trucks, assign routes, manage collection schedules, and generate performance reports.

System Administrators

System Administrators manage the technical side of the platform. Their responsibilities include managing user accounts, approving driver and officer registrations, configuring collection zones, monitoring system activity, and maintaining platform security.

2. Software Development Life Cycle (SDLC)

WasteTrack SL will be developed using the Agile Software Development Life Cycle model. Agile was selected because it supports continuous improvement, user feedback, and flexible development. This is important for a public-facing system that will be used by different groups of people, including citizens, truck drivers, and government officers with different levels of digital literacy.

Agile development is organized into short development cycles called Sprints. Each Sprint produces a working part of the system that can be tested and improved. This approach allows the project team to build the system gradually while responding to feedback from real users in Freetown communities.

2.1 Phase 1: Planning and Requirements

During this phase, the project team identifies all key stakeholders, including community residents, waste collection drivers, City Council waste management officers, Ministry officials, and System Administrators.

Stakeholder interviews and focus group discussions are conducted in communities such as Kissy, Lumley, and Congo Town. These discussions help the team understand the real waste management challenges experienced by citizens and field workers.

A Product Backlog is then created. This backlog lists all required system features based on priority, user value, and SDG impact. High-priority features include:

- Citizen waste reporting through the mobile app and USSD *111#
- Live truck tracking on an interactive map
- SMS collection schedule alerts
- Waste report management dashboard

- Driver route management
- SDG analytics dashboard

Acceptance criteria are also defined for each user story to ensure that every feature can be tested and verified.

2.2 Phase 2: System Design

In this phase, the technical structure of WasteTrack SL is designed. The system uses a three-tier client-server architecture, consisting of:

- Frontend mobile and web applications
- RESTful API backend
- Relational database

UML diagrams, including the Use Case Diagram and Class Diagram, are created to show how users interact with the system and how the main system entities are connected.

The UI/UX prototype is designed in Figma. It includes screens for onboarding, citizen dashboard, waste reporting, live truck tracking, collection schedules, driver dashboard, City Council dashboard, and SDG analytics. The USSD *111# menu flow is also designed to ensure that citizens without smartphones can still access the core services.

2.3 Phase 3: Sprint Development

Development is divided into four short Sprints. Each Sprint delivers a working part of the system.

Sprint 1: Authentication, GPS and Database Setup

This Sprint focuses on user authentication for citizens, drivers, officers, and administrators. It also includes USSD *111# registration, GPS location service integration, and PostgreSQL database schema setup.

Sprint 2: Waste Reporting and Truck Tracking

This Sprint focuses on the citizen waste report submission module, GPS-tagged photo uploads, live waste truck tracking, interactive map integration, and bin status monitoring.

Sprint 3: Alerts, USSD Integration and Dashboard

This Sprint includes SMS and push notification alerts, USSD menu integration with telecom partners, City Council dashboard development, live map monitoring, and driver route management.

Sprint 4: Analytics, Recycling Education and Testing

This Sprint focuses on the SDG analytics dashboard, recycling education content, system integration testing, performance optimization, and user acceptance testing with selected Freetown citizens and City Council officers.

Each Sprint includes daily stand-up meetings, Sprint Reviews, and Sprint Retrospectives to monitor progress and improve the development process.

2.4 Phase 4: Testing

Testing is carried out throughout the Agile process. The following testing methods are used:

Unit Testing

Unit tests are used to test individual features such as waste report submission, GPS coordinate capture, login validation, and truck capacity updates.

Integration Testing

Integration testing ensures that the mobile app, web dashboard, backend API, SMS service, map service, and database work correctly together.

User Acceptance Testing

User Acceptance Testing is conducted with real users, including Freetown community members and City Council officers. This ensures that the system is easy to use and meets real-life needs.

Performance and Security Testing

Performance testing ensures that the system can handle many users, especially during emergencies such as flooding or heavy rainfall when waste reports may increase. Security testing ensures that user information is protected through secure login, encryption, access control, and HTTPS communication.

2.5 Phase 5: Deployment

WasteTrack SL will be deployed in stages. The first deployment will be a pilot in Freetown. The system will be hosted on a cloud server and accessed through the mobile app, web dashboard, SMS, and USSD *111#.

Deployment will use a Continuous Integration and Continuous Deployment (CI/CD) pipeline managed through GitHub Actions. This will allow the team to test and release updates more efficiently.

The USSD *111# service will be integrated with local telecom partners such as Orange SL, Africell, QCell, and other licensed mobile operators to ensure wide accessibility.

2.6 Phase 6: Maintenance

After the pilot phase, the system will enter the maintenance stage. During this phase, bugs will be fixed, user feedback will be reviewed, and new features will be added.

The system may later be expanded to other districts such as Bo, Kenema, and Makeni. Monthly updates will be planned to improve system performance, update reporting requirements, and respond to the changing needs of citizens, drivers, and government stakeholders.

3. System Architecture Design

WasteTrack SL is designed using a three-tier client-server architecture. This architecture separates the system into three main layers:

1. Presentation Layer
2. Application Logic Layer
3. Data Layer

This separation makes the system easier to maintain, scale, and secure.

3.1 Architecture Overview

Tier 1: Presentation Layer

The Presentation Layer is the user-facing part of the system. It includes:

- React Native mobile application for citizens and drivers
- React.js web dashboard for City Council officers and administrators
- USSD gateway for users dialing *111#
- SMS notification service for alerts and confirmations

Tier 2: Application Logic Layer

The Application Logic Layer is handled by a Node.js REST API. It manages:

- User authentication

- Waste report processing
- GPS tagging
- Truck location updates
- Route assignment
- SMS notification dispatch
- SDG analytics generation

The API communicates with the frontend through HTTPS and connects securely to the database.

Tier 3: Data Layer

The Data Layer uses a PostgreSQL relational database. It stores:

- User profiles
- Waste reports
- Truck location logs
- Collection schedules
- Bin status data
- Recycling information
- Alerts
- SDG progress reports

The database is hosted on a secure cloud server with regular automated backups.

3.2 Architecture Components

Component	Technology	Purpose
Mobile App	React Native	Android/iOS app for citizens and drivers
Web Dashboard	React.js	Browser-based access for City Council officers and admins
USSD Gateway	*111# via Telecom API	Feature phone access for non-smartphone users
Backend API	Node.js + Express	Business logic, authentication, notifications, and analytics
Database	PostgreSQL	Persistent storage of all system data
SMS Service	Africa's Talking API	Collection alerts, report confirmations, and safety tips
Map Service	OpenStreetMap + Leaflet	Live truck tracking and waste hotspot mapping
Authentication	JWT + bcrypt	Secure token-based user sessions
Cloud Hosting	AWS / DigitalOcean	Scalable server infrastructure
CI/CD Pipeline	GitHub Actions	Automated testing and deployment

3.3 Component Interactions

The main waste reporting and collection workflow operates as follows:

- A citizen opens the mobile app or dials *111# and logs in using a phone number and PIN.
- The frontend sends the login credentials securely to the Node.js API over HTTPS.

- The API validates the user credentials against the PostgreSQL database using bcrypt password hashing and returns a JWT authentication token.
- The citizen submits a waste report with GPS coordinates, description, issue type, severity level, and optional photo evidence.
- The API stores the report in PostgreSQL and displays the location on the City Council dashboard using OpenStreetMap and Leaflet.
- A City Council officer reviews the report and assigns the nearest available waste collection truck.
- The assigned driver receives the route update through the mobile app.
- The driver's GPS location is sent back to the API in real time, allowing officers to track the truck on the live map.
- When the collection is completed, the driver marks the task as resolved.
- The system updates the report status and sends an SMS confirmation to the citizen through Africa's Talking API.
- All report and collection data is aggregated into the SDG analytics dashboard for monitoring and decision-making.

4. UML Diagrams

4.1 Use Case Diagram

The Use Case Diagram shows how the four main actors interact with WasteTrack SL.

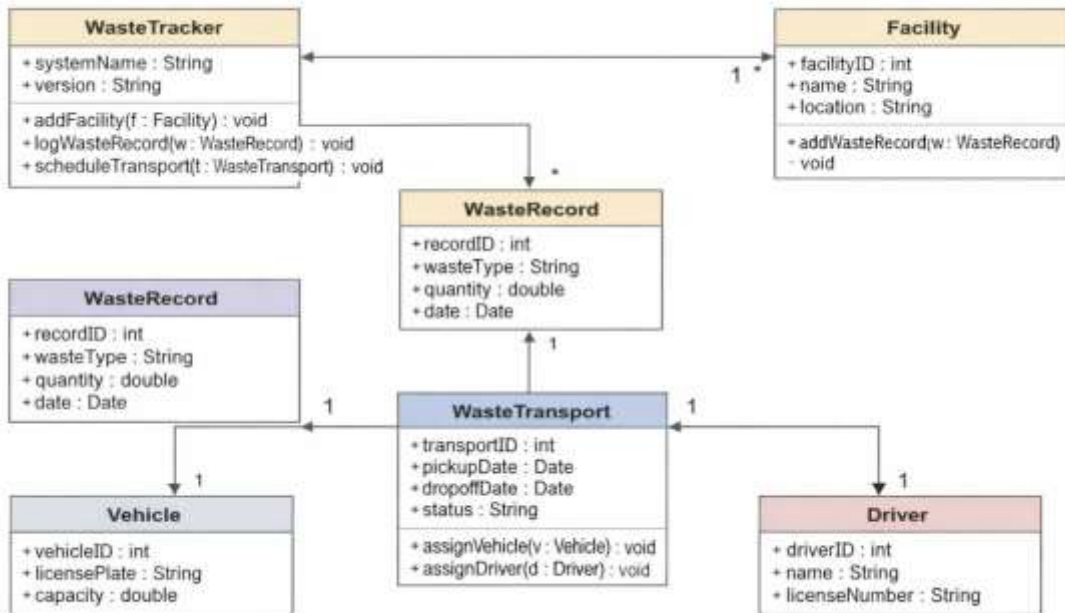


Actor	Use Cases
Citizen	Register/Login via App or USSD, Report Waste Issue, View Collection Schedule, Track Truck Live, Receive SMS Alerts, Access Recycling Tips
Waste Driver	Register/Login, View Assigned Route, Log Collection Stop, Update Truck Capacity, Report Access Problem
City Council Officer	Login, Monitor Live Map Dashboard, Manage Waste Reports, Assign Trucks, Generate SDG Reports, Broadcast Alerts
System Admin	Manage User Accounts, Approve Driver Registrations, Configure Collection Zones, Monitor System Health, View Audit Logs

4.2 Class Diagram

The Class Diagram shows the main entities in WasteTrack SL and their relationships.

WasteTrack SL Class Diagram



Class	Key Attributes	Key Methods
User	userID, name, phone, passwordHash/PINHash, role, community, location	register(), login(), updateProfile()
Citizen	reportCount, alertsEnabled, preferredLanguage	submitReport(), viewSchedule(), trackTruck()
WasteDriver	driverID, licenseNo, truckID, currentCapacity, routeID	startRoute(), logStop(), updateCapacity(), reportIssue()
CouncilOfficer	officerID, district, department	reviewReport(), assignTruck(), generateSDGReport(), broadcastAlert()

Class	Key Attributes	Key Methods
WasteReport	reportID, type, description, GPS, photoURL, status, timestamp	submit(), updateStatus(), assignToTruck(), markResolved()
TruckLocation	truckID, driverID, GPS, capacity, speed, timestamp	updateLocation(), getCapacity(), broadcastPosition()
CollectionSchedule	scheduleID, zone, date, time, truckID, type	create(), update(), notifyCitizens(), markComplete()
RecyclingTip	tipID, category, content, language	publish(), getByCategory(), sendViaSMS()
SDGReport	reportID, month, wasteCollected, coverageRate, recyclingRate	generate(), exportPDF(), sendToMinistry()

5. Prototype Development

The WasteTrack SL prototype is designed in Figma and consists of twenty-four key screens. These screens demonstrate the full user journey for Citizens, Waste Drivers, City Council Officers, System Administrators, and public users.

The prototype includes both:

- Mobile app screens: 393 × 852px
- Desktop dashboard screens: 1440 × 900px

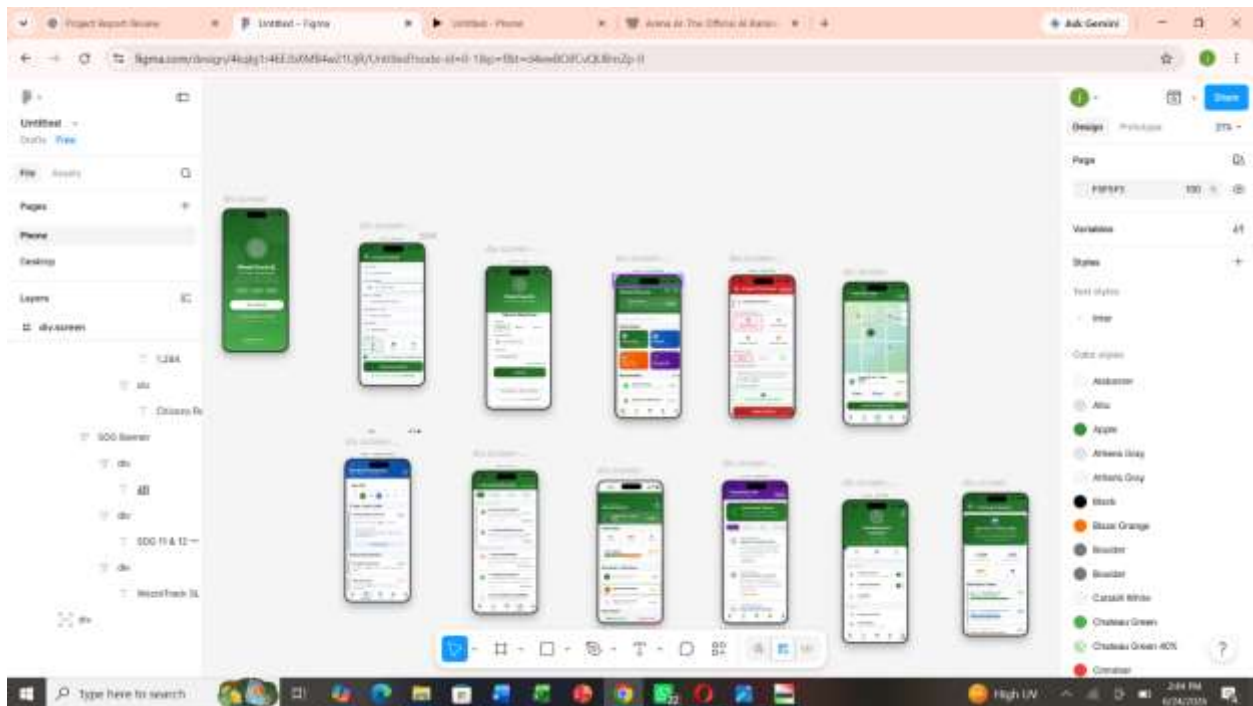
The design uses a clean and accessible visual style suitable for use in Sierra Leone's urban environment. The main color palette includes:

- Forest Green: #1B5E20 / #2E7D32 for primary actions and success states
- Dark Blue: #1565C0 for secondary actions
- Amber/Orange: #FF6F00 for warnings and important alerts
- Red: #D32F2F for urgent or danger states
- White/Light Grey: #F5F7FA for backgrounds

This color system improves readability, supports visual hierarchy, and makes the system easier to use in outdoor conditions.

5.1 Figma Screen Descriptions

Mobile Application Screen



Screen No.	Screen Name	Actor	Description
1	Splash Screen	All Users	Welcome screen showing the WasteTrack SL logo, tagline "Clean Cities. Smarter Waste.", SDG badges, and Get Started/Login options. Includes USSD access information *111#.
2	Registration	All Users	Account creation screen with role selection, full name, phone number, community, and password setup.
3	Login	All Users	Authentication screen with role selector, phone number, password field, forgot password link, and USSD alternative.
4	Citizen Home Dashboard	Citizen	Main dashboard showing greeting, next collection schedule, quick actions, and recent activity.
5	Report Waste Screen	Citizen	Form for reporting illegal dumping, overflowing bins, missed collection, or hazardous waste using GPS and photo upload.
6	Live Truck Tracker	Citizen	Map showing real-time truck location, estimated arrival time, capacity indicator, and nearby bin status.
7	Collection Schedule	Citizen	Weekly calendar showing collection days, truck assignment, waste type, and reminder options.
8	Alerts & Notifications	Citizen	Notification inbox for collection alerts, report updates, recycling announcements, and system notices.

Screen No.	Screen Name	Actor	Description
13	Landing / Hero Page	Public	Public homepage with navigation, tagline, live map preview, feature highlights, SDG section, and footer.
14	Admin Dashboard	City Council Officer	Main dashboard showing fleet KPIs, pending reports, live Freetown map, alerts, recent reports, and SDG progress.
15	Live Map View	City Council Officer	Full-screen map showing truck locations, report hotspots, bin status, and issue assignment panel.
16	Reports Management	City Council Officer	Reports table with filters, severity badges, reporter details, status tracking, photo evidence, and action buttons.
17	Fleet Manager	City Council Officer	Fleet overview showing trucks, routes, drivers, capacity, fuel, stops completed, ETA, and operational status.
18	Collection Schedule Manager	City Council Officer	Weekly calendar for scheduled collections with zone, type, truck, time, and recurring options.
19	Alerts & Notifications Centre	City Council Officer	Alert inbox with filters and broadcast form for sending SMS/app alerts to selected zones.
20	Citizens Management	System Admin	Table of registered citizens with filters, activity history, and administrative actions.

Screen No.	Screen Name	Actor	Description
21	Recycling Management	City Council Officer	Recycling dashboard showing material progress, recycling centre performance, and SDG 12 progress.
22	SDG Analytics Dashboard	City Council Officer	Analytics dashboard showing impact metrics, monthly report charts, collection trends, and SDG target progress.
23	Settings & System Configuration	System Admin	Settings page with profile update, notification preferences, and system service status indicators.
24	About & SDG Impact	Public Admin	Platform overview, SDG target summary, MIT license box, GitHub link, partners, and version information.

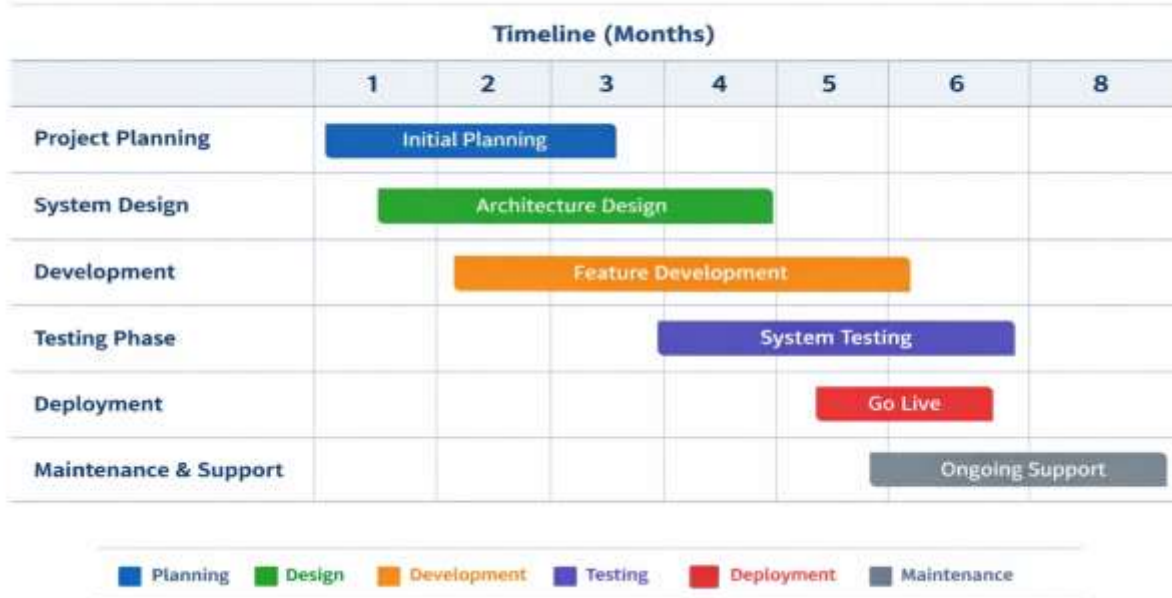
Each screen includes design annotations in the Figma file explaining the design choices, accessibility considerations, and USSD parity. Every core function available in the smartphone app has an equivalent USSD *111# menu flow so that citizens without smartphones or internet access are not excluded.

6. Project Management

6.1 Gantt Chart

The Gantt chart below outlines the project timeline across an 11-week academic development schedule.

WasteTrack SL Gantt Chart

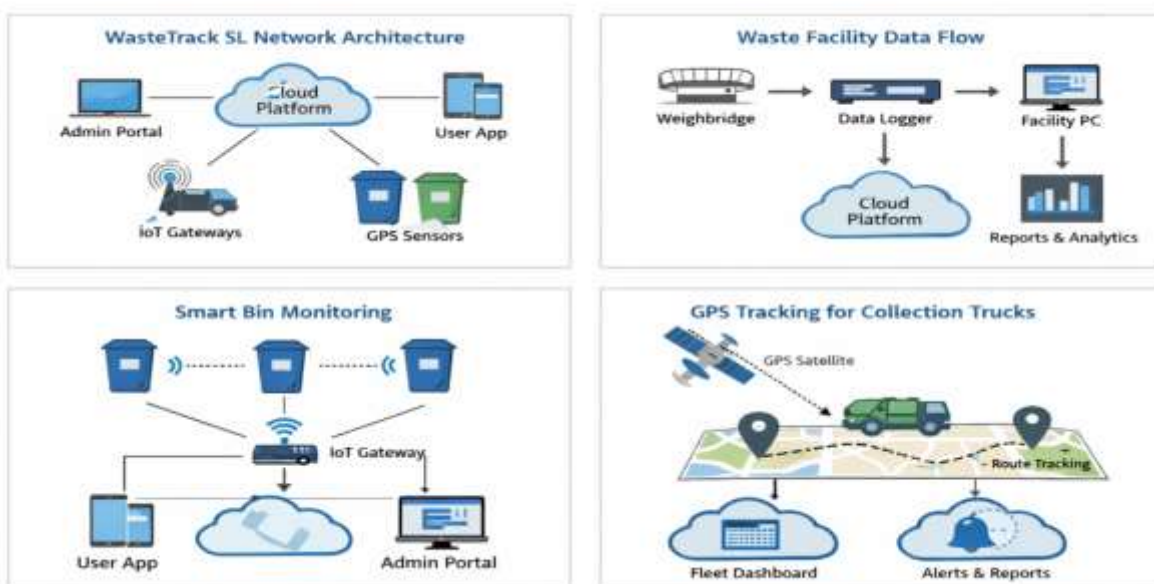


Week	Activity	Status
Week 1–2	Project Planning, Stakeholder Analysis, Requirements Gathering, SDG Alignment Mapping	Complete
Week 3–4	System Architecture Design, UML Diagrams, Figma Prototype	Complete
Week 5–6	Sprint 1: User Authentication, USSD *111#, GPS Integration, Database Schema	In Progress
Week 7	Sprint 2: Waste Report Submission, Live Truck Tracking Map, Bin Status Monitoring	Planned
Week 8	Sprint 3: SMS Alerts, USSD Menu Integration, City Council Dashboard, Driver Route Module	Planned
Week 9	Sprint 4: SDG Analytics Dashboard, Recycling Module, UAT, Performance Testing	Planned

Week	Activity	Status
Week 10	GitHub Repository Setup, API Documentation, Final Report Writing, README Completion	Planned
Week 11	Final Submission, GitHub Upload, Presentation Preparation, Examiner Review	Planned

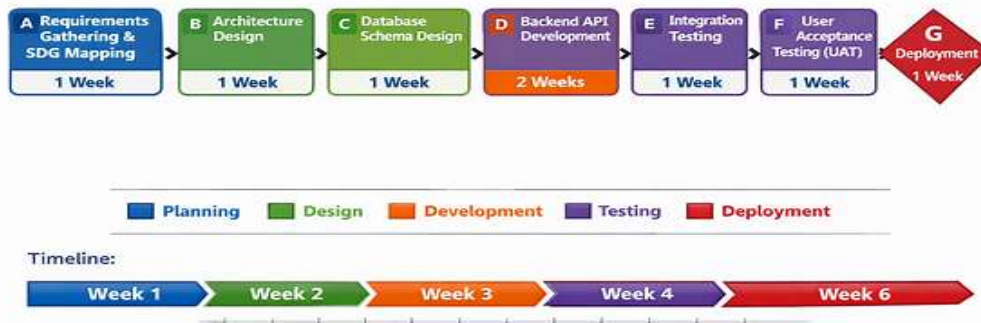
6.2 Network Diagram / Critical Path

The Network Diagram shows the sequence of major project activities and identifies the critical path.



The critical path for WasteTrack SL is:

WasteTrack SL – Critical Path Diagram



Any delay in these activities may directly affect the final submission deadline. Other tasks, such as Figma prototype design and USSD flow design, can run in parallel and support the development process.

Activity	Duration	Depends On	Critical?
A: Requirements Gathering & SDG Mapping	1 week	—	Yes
B: System Architecture Design	1 week	A	Yes
C: Database Schema Design	1 week	B	Yes
D: Figma Prototype Design	1 week	B	No
E: USSD *111# Flow Design	1 week	B	No
F: Sprint 1-Auth, GPS & Database	2 weeks	C	Yes
G: Sprint 2-Waste Reporting & Tracking	1 week	F, D	Yes
H: Sprint 3-Alerts, Dashboard & Driver Module	1 week	G, E	Yes
I: Sprint 4-Analytics, Testing & UAT	1 week	H	Yes
J: UAT with Real Freetown Users	1 week	I	Yes

Activity	Duration	Depends On	Critical?
K: GitHub Upload & Documentation	1 week	I	No
L: Final Submission	—	J, K	Yes

7. License, Design Quality and SDG Relevance

7.1 Software License

WasteTrack SL is licensed under the MIT License. This license was chosen because it supports open-source sharing, collaboration, and adaptation.

The MIT License allows individuals, governments, NGOs, and developers to freely use, copy, modify, publish, distribute, and adapt the software, as long as the original copyright notice and license text are included.

As a Digital Public Good, WasteTrack SL must remain open and accessible so that it can be reused in Sierra Leone and other countries facing similar waste management problems. The MIT License also aligns with the Digital Public Goods Standard, which encourages open-source licensing for public-interest digital solutions.

External resources used in the system are properly attributed in the documentation and README file. These include:

- Africa's Talking API
- OpenStreetMap
- React Native
- Leaflet.js

MIT License Summary

Copyright (c) 2026 WasteTrack SL.

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files, to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the software, provided that the copyright notice and permission notice are included in all copies.

The full MIT License text will be included in the GitHub repository under the LICENSE file.

7.2 Design Quality / UI/UX

The WasteTrack SL interface is designed with the following quality standards:

Accessibility

All screens are designed to meet WCAG 2.1 AA contrast requirements. Mobile text uses a minimum size of 16sp. Icons are supported with text labels to help users with low literacy levels. The interface is also designed to remain readable in outdoor environments with bright sunlight.

Consistency

The same design system is used across all 24 Figma screens. This includes consistent typography, spacing, buttons, cards, navigation bars, and status badges. The main font used is Inter, and the interface follows an 8px spacing grid.

Navigation

The citizen mobile app uses a simple bottom navigation bar with five main items:

- Home
- Report
- Track
- Alerts
- Profile

The driver interface is simplified to reduce distraction during fieldwork. It focuses on route information, truck capacity, and issue reporting.

USSD Parity

Every core smartphone feature has a USSD *111# equivalent. This means citizens without smartphones or internet access can still:

- Report waste problems
- Check collection schedules
- Receive safety tips
- Get SMS confirmations

This ensures that WasteTrack SL is inclusive and accessible to all citizens, regardless of device type or income level.

7.3 SDG Relevance

WasteTrack SL supports several Sustainable Development Goals.

SDG 11: Sustainable Cities and Communities

WasteTrack SL directly supports Target 11.6, which focuses on reducing the environmental impact of cities, especially through improved waste management.

The system helps the Freetown City Council respond faster to illegal dumping, overflowing bins, and blocked drainage areas. Live truck tracking and route planning can improve collection efficiency and reduce uncollected waste in public spaces.

SDG 12: Responsible Consumption and Production

WasteTrack SL supports Target 12.4 and Target 12.5 by encouraging proper waste handling, recycling, reuse, and responsible disposal.

The recycling education module teaches citizens how to separate waste, compost organic materials, and participate in recycling drives. The system also tracks recycling data by material type, helping authorities monitor progress.

SDG 3: Good Health and Well-being

WasteTrack SL also supports Target 3.9, which aims to reduce illness caused by pollution and unsafe environments. By helping remove waste from public spaces and drainage areas, the system can reduce disease risks such as malaria, cholera, and other sanitation-related illnesses.

SDG 6: Clean Water and Sanitation

Waste blocking drainage systems contributes to flooding and water contamination. By allowing citizens to report blockages and enabling faster response, WasteTrack SL supports cleaner waterways and safer sanitation conditions.

WasteTrack SL also supports the Government of Sierra Leone's environmental priorities and the Freetown City Council's Transform Freetown initiative, which identifies waste management as a key part of sustainable urban development.

8. GitHub Repository

A GitHub repository has been created for WasteTrack SL to support version control, collaboration, documentation, and open-source access.

8.1 Repository Structure

Folder / File	Contents
/docs	Final project report, system design rationale, and user manual
/diagrams	Use Case Diagram, Class Diagram, Architecture Diagram, Gantt Chart, and Network Diagram
/prototype	Figma export screenshots of all 24 screens, organized by actor type
/src	Backend source code, API files, database schema, routes, and environment template
README.md	Project overview, setup guide, API documentation, SDG alignment, MIT license, and deployment guide
LICENSE	Full MIT License text

8.2 Collaboration

The repository is set to Public to comply with the MIT open-source license and the Digital Public Goods Standard.

The course examiner has been added as a collaborator with read access for assessment purposes. Commits are made using clear and descriptive messages following conventional commit standards, such as:

- feat: add live truck GPS tracking endpoint
- fix: validate waste report location input
- docs: update README with USSD flow documentation

Branch management follows the GitFlow approach:

- main branch contains stable deployable code

- develop branch integrates Sprint features
- Feature branches are used for individual modules, such as feature/waste-report-form and feature/sms-alert-service

8.3 Backend Source Code

The backend source code is stored in the /src folder. It is built using Node.js and Express.js and follows RESTful API design principles.

The backend contains the following route modules:

citizens.js

Handles citizen and driver registration, login, JWT token generation, profile management, and role-based access control.

reports.js

Manages waste report submission, GPS coordinate validation, photo URL storage, report tracking, and officer assignment.

trucks.js

Receives real-time GPS updates from driver mobile apps, stores truck location logs, manages truck capacity, and supports live map tracking.

schedules.js

Manages collection schedule creation, zone configuration, citizen notifications, and schedule completion records.

alerts.js

Integrates with Africa's Talking API to send SMS alerts for collection reminders, report updates, and emergency waste notices.

The PostgreSQL database schema creates the following tables:

- Users
- Citizens
- Drivers
- WasteReports
- TruckLocations
- CollectionSchedules
- RecyclingTips
- Alerts
- SDGReports

Key security measures include:

- bcrypt hashing for user passwords and PINs
- JWT authentication with token expiry
- Input validation and sanitization
- HTTPS-only communication
- Role-based access control
- Secure storage of sensitive user data

9. Conclusion

WasteTrack SL shows how Digital Public Goods can be used to solve real problems in communities while supporting global development goals. The system is designed to improve waste reporting, truck tracking, collection scheduling, recycling education, and data-driven decision-making in Freetown.

By combining a mobile app, web dashboard, SMS alerts, and USSD *111# access, WasteTrack SL directly addresses three major challenges in Freetown's waste management system: the reporting gap, the operational gap, and the data gap.

The system uses the Agile SDLC model, allowing continuous feedback from citizens, waste drivers, and City Council officers. Its three-tier architecture, built with React Native, Node.js, and PostgreSQL, provides a strong foundation for future growth. The USSD feature ensures that citizens without smartphones or internet access are not left behind.

The MIT open-source license allows WasteTrack SL to be freely reused and adapted by governments, NGOs, and developers in Sierra Leone and beyond. Future work will include piloting the platform in communities such as Kissy and Lumley, partnering with the Freetown City Council, and integrating the system with national environmental reporting structures to support Sierra Leone's SDG 11 and SDG 12 commitments.

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